

## LF synthetic fixture results

PyMuPDF 1.28.2; ARPipe (`arpipe/triage.py`) imported unmodified from main @ 52ff2fe. Synthetic PDFs only — no ARPipe corpus PDF was used.

CER = normalised Levenshtein distance vs ground truth of what a reader sees on the page (lower is better). R1 = ignore /ToUnicode, map glyph IDs back through the embedded font's own `cmap`. R3 = apply the fixture's ASCII-slot table blindly to all text. Font-level flag = per-font junk ratio >2% or English-lexicon hit rate <0.3.

| Fixture                        | Mechanism                                                                      | ARPipe kind →<br>route   | mojibake<br>ratio | Native<br>CER | Tesseract<br>CER   | R1<br>cmap<br>CER                         | R3<br>blind<br>table<br>CER | Font-<br>level<br>flag |
|--------------------------------|--------------------------------------------------------------------------------|--------------------------|-------------------|---------------|--------------------|-------------------------------------------|-----------------------------|------------------------|
| F00_clean                      | healthy<br>control<br>(subset font)                                            | digital →<br>NATIVE_READ | 0.0               | 0.0           | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.517                       | False                  |
| F00b_clean_fullfont            | healthy<br>control (full<br>font)                                              | digital →<br>NATIVE_READ | 0.0               | 0.0           | 0.0 (eng)          | 0.0                                       | 0.517                       | False                  |
| F01_missing_tounicode_subset   | /ToUnicode<br>missing,<br>Identity-H,<br>subset                                | broken_text →<br>OCR     | 0.15629           | 0.989         | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.989                       | True                   |
| F02_missing_tounicode_fullfont | /ToUnicode<br>missing,<br>Identity-H,<br>full font                             | broken_text →<br>OCR     | 0.15629           | 0.989         | 0.0 (eng)          | 0.0                                       | 0.989                       | True                   |
| F03_wrong_tounicode_shift      | /ToUnicode<br>present but<br>wrong<br>(substitution)                           | digital →<br>NATIVE_READ | 0.0               | 0.828         | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.831                       | True                   |
| F03b_wrong_tounicode_fullfont  | same, full<br>font (cmap<br>retained)                                          | digital →<br>NATIVE_READ | 0.0               | 0.828         | 0.0 (eng)          | 0.0                                       | 0.831                       | True                   |
| F04_pua_tounicode              | /ToUnicode<br>maps to<br>Private Use<br>Area                                   | digital →<br>NATIVE_READ | 0.0               | 0.989         | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.989                       | True                   |
| F05_ascii_slot_legacy_deva     | ASCII-slot<br>'legacy'<br>Devanagari<br>font (Kruti-<br>Dev-type<br>mechanism) | digital →<br>NATIVE_READ | 0.0               | 0.716         | 0.093<br>(hin+eng) | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.0                         | True                   |

| Fixture                              | Mechanism                                                                                                                      | ARPipe kind →<br>route                                        | mojibake<br>ratio | Native<br>CER | Tesseract<br>CER   | R1<br>cmap<br>CER                         | R3<br>blind<br>table<br>CER | Font-<br>level<br>flag |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------|-------------------|---------------|--------------------|-------------------------------------------|-----------------------------|------------------------|
| F06_mixed_heading_missing_tu         | clean body +<br>heading font<br>with missing<br>/ToUnicode                                                                     | digital →<br>NATIVE_READ                                      | 0.00465           | 0.048         | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.54                        | True                   |
| F07_mixed_heading_wrong_tu           | clean body +<br>heading font<br>with wrong<br>/ToUnicode                                                                       | digital →<br>NATIVE_READ                                      | 0.0               | 0.043         | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.535                       | True                   |
| F08_mixed_heading_legacy_slot        | clean body +<br>ASCII-slot<br>Devanagari<br>heading                                                                            | digital →<br>NATIVE_READ                                      | 0.0               | 0.016         | 0.004<br>(hin+eng) | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.506                       | True                   |
| F09_bilingual_half_legacy            | bilingual<br>page: ½<br>English, ½<br>ASCII-slot<br>Hindi                                                                      | digital →<br>NATIVE_READ                                      | 0.0               | 0.271         | 0.036<br>(hin+eng) | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.319                       | True                   |
| F10_image_plus_wrong_invisible_layer | page image +<br>invisible (Tr<br>3) wrong text<br>layer                                                                        | digital →<br>NATIVE_READ                                      | 0.0               | 0.828         | 0.0 (eng)          | n/a<br>(subset<br>font<br>has no<br>cmap) | 0.831                       | True                   |
| F11_unicode_deva_shaped              | valid Unicode<br>Devanagari<br>font,<br>HarfBuzz-<br>shaped<br>(conjuncts,<br>reph, pre-<br>base matra)<br>— no legacy<br>font | digital →<br>NATIVE_READ<br>(script = mixed,<br>mojibake 0.0) | 0.0               | 0.298         | 0.0 (hin)          | not run                                   | not<br>run                  | not<br>run             |

Notes

- F01/F02 are caught by ARPipe only because the space glyph has glyph ID 3, so each raw-CID space becomes control character U+0003. Detection of missing /ToUnicode therefore depends on incidental glyph numbering, not on a mapping check.
- Subset fonts written by MuPDF carry no `cmap` and no glyph names (`post` dropped), so R1 and glyph-name checks are impossible on them; on full fonts R1 restores the text exactly (F02, F03b) and the /ToUnicode-vs-cmap agreement falls from 1.0 (F00b) to 0.186 (F03b).
- R3 (a legacy table) gives 0.0 on F05 only because the table is the fixture's own; applied blindly to a clean page it corrupts it (CER 0.517).
- The MD&A heading regex (`patterns.MDA_HEADING_RE`) matches the clean page but not F06/F07, both routed `NATIVE_READ`.
- pypdf 3.17.4 raised an exception on MuPDF's /ToUnicode (it writes 5-hex-digit destinations for astral code points, not UTF-16 pairs); pypdf CERs in `results.json` are harness errors, not measurements. `pdfminer.six` raised on the full-font F02.
- Font-level flag: 0/2 false positives on clean fixtures — a sample of two, not an FP-rate estimate.